Ref. [29] — Global Convergence Using Optimal Transport

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1. The problem: why can non-convex particle training find a global solution?

In a wide two-layer network, output is a sum of many nonlinear neuron features. Optimizing finite neuron locations and amplitudes is non-convex. Chizat and Bach reformulate the task in terms of an *unknown measure*: once the distribution of neurons is the optimization variable, the objective can become convex even while its finite-particle parameterization remains non-convex.

2. The exact modeling idea

Their broad formulation is a convex loss of a feature mixture plus a convex measure regularizer,

$$J(\mu) = R\left(\int_\Theta \phi(\theta)\, d\mu(\theta)\right) + G(\mu).$$

Θ is the neuron/particle parameter space; $\phi(\theta)$ is the feature represented by a neuron; μ is a (possibly signed) measure over neurons; and R is convex in the *resulting predictor*. If μ is represented by finitely many particles, then

$$\mu_m = \sum_{j=1}^m a_j \delta_{\theta_j}, \qquad \int \phi\, d\mu_m = \sum_{j=1}^m a_j \phi(\theta_j).$$

This recovers finite-width feature sums, such as single-hidden-layer network models, but the parameters $\{a_j, \theta_j\}$ still enter nonlinearly.

3. Why optimal transport appears

The method follows the continuous-time evolution of particle masses and positions. In the many-particle limit, those trajectories are described by a *Wasserstein gradient flow*, an evolution of a distribution rather than a single vector. A schematic continuity equation is

$$\partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0, \quad v_t \propto -\nabla_\theta \frac{\delta J}{\delta \mu}(\theta),$$

where $\delta J/\delta \mu$ is a first variation of the objective. This is explanatory notation: the precise formulation depends on the measure model, homogeneity and regularizer used in the article.

Actual contribution. Under specified regularity, homogeneity and initialization/support assumptions, the appropriate infinite-particle gradient flow converges to global minimizers of the convex-in-measure objective. The paper also reports numerical evidence that the many-particle picture is informative at moderate finite width.

4. Why this is not just the neural tangent kernel

NTK analysis often keeps features nearly frozen around random initialization. In mean-field/transport regimes, particles can move nontrivially in parameter space, thereby changing the learned feature distribution. Thus "wide network" alone is insufficient to identify which theory applies: the limiting scaling and initialization matter.

5. Assumptions, qualifications and implications

The guarantee concerns the relevant *continuous-time many-particle limit* and its hypotheses. It is not a blanket theorem that arbitrary finite ReLU/Transformer networks trained by Adam at large learning rates reach global minima. Nor does convexity in μ by itself make the finite particle objective convex in (a_j, θ_j).

Survey relevance (RQ4). This is a second route to favorable optimization, distinct from explicit saddle escape. Its price is a measure-valued limit plus support/initialization assumptions. Contrast with lazy NTK convergence and with saddle-to-saddle dynamics at small initialization.

6. Our notes

The useful insight is that a neural network can be non-convex in the individual neurons but convex when the whole neuron distribution is the variable. The authors show global behavior in an idealized infinitely wide particle flow; they do not say that every practical finite network behaves like this. **Original source:** https://papers.neurips.cc/paper_files/paper/2018/hash/a1afc58c6ca9540d057299ec3016d726-Abstract.html.